1. A silicon diode has $V = 0.7\ \mathrm{V}$ and $I = 10\ \mathrm{mA}$. Find the static resistance.

Formula:

$$R = \frac{V}{I}$$

Given:

$$V = 0.7\ \mathrm{V}, \quad I = 10\ \mathrm{mA} = 0.01\ \mathrm{A}$$

So,

$$R = \frac{0.7}{0.01} = 70\ \Omega$$

Answer: $70\ \Omega$

2. If the forward voltage changes from 0.68 V to 0.72 V and current changes from 8 mA to 18 mA, find dynamic resistance.

Formula:

$$r_d = \frac{\Delta V}{\Delta I}$$

Given:

$$\Delta V = 0.72 - 0.68 = 0.04\ \mathrm{V}$$

$$\Delta I = 18\ \mathrm{mA} - 8\ \mathrm{mA} = 10\ \mathrm{mA} = 0.01\ \mathrm{A}$$

So,

$$r_d = \frac{0.04}{0.01} = 4\ \Omega$$

Answer: $4\ \Omega$

3. A diode in reverse bias has $V_R = 10\ \mathrm{V}$ and $I_R = 2\ \mu A$. Find reverse static resistance.

Formula:

$$R = \frac{V}{I}$$

Given:

$$V = 10\ \mathrm{V}, \quad I = 2\mu A = 2 \times 10^{-6}\ \mathrm{A}$$

So,

$$R = \frac{10}{2 \times 10^{-6}} = 5 \times 10^{6}\ \Omega$$

$$R = 5\ \mathrm{M}\Omega$$

Answer: $5\ \mathrm{M}\Omega$

4. For a diode, $V_f = 0.75\ \text{V}, I_f = 15\ \text{mA}$. Calculate power dissipated.

Formula:

$$P = VI$$

Given:

$$V = 0.75\ \text{V}, \quad I = 15\ \text{mA} = 0.015\ \text{A}$$

So,

$$P = 0.75 \times 0.015 = 0.01125\ \text{W}$$

$$P = 11.25\ \text{mW}$$

Answer: $11.25\ \text{mW}$

5. The reverse current of a diode is $5\ \mu A$ at 20 V. Find resistance.

Formula:

$$R = \frac{V}{I}$$

Given:

$$V = 20\ \text{V}, \quad I = 5\mu A = 5 \times 10^{-6}\ \text{A}$$

So,

$$R = \frac{20}{5 \times 10^{-6}} = 4 \times 10^{6}\ \Omega$$

$$R = 4\ \text{M}\Omega$$

Answer: $4\ \text{M}\Omega$

6. If the knee voltage of a silicon diode is approximately 0.7 V, what is the expected conduction state at 0.5 V?

Since

$$0.5\ \text{V} < 0.7\ \text{V}$$

the diode is **not yet fully turned on.**

Answer: The diode is OFF or only very slightly conducting.

7. A Zener diode has breakdown voltage 5.6 V. If reverse voltage is 6 V, in which region is it operating?

Since

$$6\ \text{V} > 5.6\ \text{V}$$

the Zener is beyond breakdown voltage.

Answer: It is operating in the reverse breakdown region.

8. A Zener diode maintains 5.6 V across it while current changes from 10 mA to 20 mA and voltage changes from 5.58 V to 5.62 V. Find dynamic resistance.

Formula:

$$r_z = \frac{\Delta V}{\Delta I}$$

Given:

$$\Delta V = 5.62 - 5.58 = 0.04 \text{ V}$$

$$\Delta I = 20 \text{ mA} - 10 \text{ mA} = 10 \text{ mA} = 0.01 \text{ A}$$

So,

$$r_z = \frac{0.04}{0.01} = 4\ \Omega$$

Answer: $4\ \Omega$

9. In forward bias, current is 20 mA and voltage is 0.8 V. Find static resistance.

Formula:

$$R = \frac{V}{I}$$

Given:

$$V = 0.8 \text{ V}, \quad I = 20 \text{ mA} = 0.02 \text{ A}$$

So,

$$R = \frac{0.8}{0.02} = 40\ \Omega$$

Answer: $40\ \Omega$

10. If a diode carries 25 mA at 0.75 V, find the power.

Formula:

$$P = VI$$

Given:

$$V = 0.75 \text{ V}, \quad I = 25 \text{ mA} = 0.025 \text{ A}$$

So,

$$P = 0.75 \times 0.025 = 0.01875 \text{ W}$$

$$P = 18.75 \text{ mW}$$

Answer: 18.75 mW

11. If a Zener diode has $V_Z = 5.6\ \mathrm{V}$ and current $I_Z = 12\ \mathrm{mA}$, find power dissipation.

Formula:

$$P = VI$$

Given:

$$V = 5.6\ \mathrm{V}, \quad I = 12\ \mathrm{mA} = 0.012\ \mathrm{A}$$

So,

$$P = 5.6 \times 0.012 = 0.0672\ \mathrm{W}$$

$$P = 67.2\ \mathrm{mW}$$

Answer: $67.2\ \mathrm{mW}$

12. A diode's voltage changes by 0.1 V while current changes by 25 mA. Find dynamic resistance.

Formula:

$$r_d = \frac{\Delta V}{\Delta I}$$

Given:

$$\Delta V = 0.1\ \mathrm{V}$$

$$\Delta I = 25\ \mathrm{mA} = 0.025\ \mathrm{A}$$

So,

$$r_d = \frac{0.1}{0.025} = 4\ \Omega$$

Answer: $4\ \Omega$

13. If reverse saturation current is $3\ \mu A$ and reverse voltage is 15 V, calculate reverse resistance.

Formula:

$$R = \frac{V}{I}$$

Given:

$$V = 15\ \mathrm{V}, \quad I = 3\mu A = 3 \times 10^{-6}\ \mathrm{A}$$

So,

$$R = \frac{15}{3 \times 10^{-6}} = 5 \times 10^{6}\ \Omega$$

$$R = 5\ \mathrm{M}\Omega$$

Answer: $5\ \mathrm{M}\Omega$

14. For two measured points of a diode: $(0.65\ \mathrm{V}, 5\ \mathrm{mA})$ and $(0.70\ \mathrm{V}, 15\ \mathrm{mA})$, find dynamic resistance.

Formula:

$$r_d = \frac{\Delta V}{\Delta I}$$

Given:

$$\Delta V = 0.70 - 0.65 = 0.05\ \mathrm{V}$$

$$\Delta I = 15\ \mathrm{mA} - 5\ \mathrm{mA} = 10\ \mathrm{mA} = 0.01\ \mathrm{A}$$

So,

$$r_d = \frac{0.05}{0.01} = 5\ \Omega$$

Answer: $5\ \Omega$

15. A Zener diode works at 5.6 V and 30 mA. Find power rating needed.

Formula:

$$P = VI$$

Given:

$$V = 5.6\ \mathrm{V}, \quad I = 30\ \mathrm{mA} = 0.03\ \mathrm{A}$$

So,

$$P = 5.6 \times 0.03 = 0.168\ \mathrm{W}$$

$$P = 168\ \mathrm{mW}$$

Answer: Minimum power rating needed = $0.168\ \mathrm{W}$

In practice, choose a higher standard rating, such as **0.25 W**.

1. For a half-wave rectifier, if $V_{peak} = 10\ \text{V}$, find V_{DC}.

Formula for half-wave rectifier:

$$V_{DC} = 0.318 V_{peak}$$

Substitute:

$$V_{DC} = 0.318 \times 10 = 3.18\ \text{V}$$

Answer: $3.18\ \text{V}$

2. If $V_{peak} = 12\ \text{V}$, calculate the average DC output of half-wave rectifier.

$$V_{DC} = 0.318 V_{peak}$$

$$V_{DC} = 0.318 \times 12 = 3.816\ \text{V}$$

Answer: $3.816\ \text{V}$

3. If the input is $8\ \text{V}_{p-p}$, find V_{peak}.

Formula:

$$V_{peak} = \frac{V_{p-p}}{2}$$

$$V_{peak} = \frac{8}{2} = 4\ \text{V}$$

Answer: $4\ \text{V}$

4. For $8\ \text{V}_{p-p}$, calculate V_{DC} of a half-wave rectifier.

First find peak voltage:

$$V_{peak} = \frac{8}{2} = 4\ \text{V}$$

Now:

$$V_{DC} = 0.318 \times 4 = 1.272\ \text{V}$$

Answer: $1.272\ \text{V}$

5. If $R_L = 10k\Omega$ and $V_{DC} = 3.18\ \text{V}$, find I_{DC}.

Formula:

$$I_{DC} = \frac{V_{DC}}{R_L}$$

$$I_{DC} = \frac{3.18}{10000} = 0.000318\ \text{A}$$

$$I_{DC} = 0.318\ \text{mA}$$

Answer: $0.318\ \text{mA}$

6. If $V_{peak} = 15\ \text{V}$, what is the PIV required for half-wave rectifier diode?

For a half-wave rectifier:

$$PIV = V_{peak}$$

So,

$$PIV = 15\ \text{V}$$

Answer: $15\ \text{V}$

7. A silicon diode drop is $0.7\ \text{V}$. If input peak is $5\ \text{V}$, estimate output peak.

For one conducting diode:

$$V_{out(peak)} = V_{in(peak)} - V_D$$

$$V_{out(peak)} = 5 - 0.7 = 4.3\ \text{V}$$

Answer: $4.3\ \text{V}$

8. If output DC voltage is $2.54\ \text{V}$, estimate V_{peak}.

For half-wave rectifier:

$$V_{DC} = 0.318 V_{peak}$$

So,

$$V_{peak} = \frac{V_{DC}}{0.318}$$

$$V_{peak} = \frac{2.54}{0.318} \approx 7.99\ \text{V}$$

Answer: $\approx 8\ \text{V}$

9. For a full-wave rectifier, if $V_{peak} = 10\ \text{V}$, find V_{DC}.

Formula:

$$V_{DC} = 0.636 V_{peak}$$

$$V_{DC} = 0.636 \times 10 = 6.36\ \text{V}$$

Answer: $6.36\ \text{V}$

10. If $V_{peak} = 12\ \text{V}$, calculate average DC output of full-wave rectifier.

$$V_{DC} = 0.636 \times 12 = 7.632\ \text{V}$$

Answer: $7.632\ \text{V}$

11. Compare DC outputs of half-wave and full-wave rectifiers for $V_{peak} = 10\ \text{V}$.

Half-wave:

$$V_{DC(HW)} = 0.318 \times 10 = 3.18\ \text{V}$$

Full-wave:

$$V_{DC(FW)} = 0.636 \times 10 = 6.36\ \text{V}$$

So the full-wave output is twice the half-wave output.

Answer:

- Half-wave DC output = **3.18 V**
- Full-wave DC output = **6.36 V**

12. If $V_{DC} = 6.36\ \text{V}$ and $R_L = 10k\Omega$, find I_{DC}.

$$I_{DC} = \frac{V_{DC}}{R_L}$$

$$I_{DC} = \frac{6.36}{10000} = 0.000636\ \text{A}$$

$$I_{DC} = 0.636\ \text{mA}$$

Answer: $0.636\ \text{mA}$

13. In a bridge rectifier, if each conducting diode drops $0.7\ \text{V}$, total drop in one conduction path is how much?

In a bridge rectifier, current passes through **2 diodes** in each half-cycle.

So:

$$V_{drop} = 0.7 + 0.7 = 1.4\ \text{V}$$

Answer: $1.4\ \text{V}$

14. If input peak is $12\ \text{V}$, estimate output peak of a bridge rectifier considering two diode drops.

$$V_{out(peak)} = V_{in(peak)} - 2V_D$$

$$V_{out(peak)} = 12 - 1.4 = 10.6\ \text{V}$$

Answer: $10.6\ \text{V}$

15. If input frequency is $50\ \text{Hz}$, what is ripple frequency of full-wave rectifier?

For full-wave rectifier:

$$f_{ripple} = 2f_{input}$$

$$f_{ripple} = 2 \times 50 = 100\ \text{Hz}$$

Answer: $100\ \text{Hz}$

16. If input frequency is $1\ \mathrm{kHz}$, what is ripple frequency of full-wave rectifier?

$$f_{ripple} = 2f_{input}$$

$$f_{ripple} = 2 \times 1000 = 2000\ \mathrm{Hz}$$

$$f_{ripple} = 2\ \mathrm{kHz}$$

Answer: $2\ \mathrm{kHz}$

17. If load current is $2\ \mathrm{mA}$ and load resistance is $10k\Omega$, find output DC voltage.

Formula:

$$V = IR$$

$$V = 2\ \mathrm{mA} \times 10k\Omega$$

Convert:

$$I = 0.002\ \mathrm{A}, \quad R = 10000\Omega$$

$$V = 0.002 \times 10000 = 20\ \mathrm{V}$$

Answer: $20\ \mathrm{V}$

18. If half-wave ripple factor is 1.21 and full-wave ripple factor is 0.482, which one is better and why?

Lower ripple factor means smoother DC output.

Since:

$$0.482 < 1.21$$

the full-wave rectifier is better.

Answer: Full-wave rectifier is better because it has lower ripple factor and produces smoother DC output.

19. Calculate power delivered to $R_L = 10k\Omega$ when $V_{DC} = 5\ \mathrm{V}$.

Formula:

$$P = \frac{V^2}{R}$$

$$P = \frac{5^2}{10000} = \frac{25}{10000} = 0.0025\ \mathrm{W}$$

$$P = 2.5\ \mathrm{mW}$$

Answer: $2.5\ \mathrm{mW}$

20. If V_{ripple} decreases after using a 2200 μF capacitor, what happens to the smoothness of DC output?

When ripple decreases, the DC output becomes more steady and smooth.

Answer: The output becomes smoother and closer to pure DC.

1. If $V_{CC} = 12\ \text{V}$ and collector resistor $R_C = 4.7k\Omega$, find maximum collector current assuming $V_{CE} = 0$.

Formula:

$$I_C = \frac{V_{CC}}{R_C}$$

Given:

$$V_{CC} = 12\ \text{V}, \quad R_C = 4.7k\Omega = 4700\Omega$$

So,

$$I_C = \frac{12}{4700} = 0.002553\ \text{A}$$

$$I_C \approx 2.55\ \text{mA}$$

Answer: $2.55\ \text{mA}$

2. If $V_{CC} = 12\ \text{V}, R_C = 4.7k\Omega$, and $V_{CE(sat)} = 0.2\ \text{V}$, find I_C.

Formula:

$$I_C = \frac{V_{CC} - V_{CE(sat)}}{R_C}$$

Substitute:

$$I_C = \frac{12 - 0.2}{4700} = \frac{11.8}{4700} = 0.002511\ \text{A}$$

$$I_C \approx 2.51\ \text{mA}$$

Answer: $2.51\ \text{mA}$

3. If $V_{BB} = 5\ \text{V}, R_B = 10k\Omega$, and $V_{BE} = 0.7\ \text{V}$, calculate I_B.

Formula:

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

Substitute:

$$I_B = \frac{5 - 0.7}{10000} = \frac{4.3}{10000} = 0.00043\ \text{A}$$

$$I_B = 0.43\ \text{mA}$$

Answer: $0.43\ \text{mA}$

4. If $I_B = 0.43\ \text{mA}$ and $\beta = 100$, estimate collector current in active region.

Formula:

$$I_C = \beta I_B$$

Substitute:

$$I_C = 100 \times 0.43\ \text{mA} = 43\ \text{mA}$$

Answer: $43\ \text{mA}$

Note: In an actual switch circuit, this current may be limited by R_C and supply voltage.

5. If transistor is OFF, what are approximate values of I_B, I_C, and V_{CE}?

In cutoff region:

- base current is nearly zero
- collector current is nearly zero
- collector-emitter voltage is approximately equal to supply voltage

So,

$$I_B \approx 0$$

$$I_C \approx 0$$

$$V_{CE} \approx V_{CC}$$

If $V_{CC} = 12\ \text{V}$, then:

$$V_{CE} \approx 12\ \text{V}$$

Answer:

$$I_B \approx 0, \quad I_C \approx 0, \quad V_{CE} \approx 12\ \text{V}$$

6. If transistor is saturated and $I_C = 2.5\ \text{mA}$, $V_{CE} = 0.2\ \text{V}$, find transistor power.

Formula:

$$P = V_{CE} \times I_C$$

Convert current:

$$I_C = 2.5\ \text{mA} = 0.0025\ \text{A}$$

Now,

$$P = 0.2 \times 0.0025 = 0.0005\ \text{W}$$

$$P = 0.5\ \text{mW}$$

Answer: $0.5\ \text{mW}$

7. Determine the two intercepts of DC load line for $V_{CC} = 12\ \text{V}$ and $R_C = 4.7k\Omega$.

Load line equation:

$$V_{CC} = I_C R_C + V_{CE}$$

Intercept 1: when $I_C = 0$

$$V_{CE} = V_{CC} = 12\ \text{V}$$

So one intercept is:

$$(V_{CE} = 12\ \text{V},\ I_C = 0)$$

Intercept 2: when $V_{CE} = 0$

$$I_C = \frac{V_{CC}}{R_C} = \frac{12}{4700} = 2.55\ \text{mA}$$

So the second intercept is:

$$(V_{CE} = 0,\ I_C = 2.55\ \text{mA})$$

Answer:

- $(12\ \text{V},\ 0)$
- $(0,\ 2.55\ \text{mA})$

8. If $V_i = 10\ \text{mV}$ and $V_o = 1\ \text{V}$, find voltage gain A_v.

Formula:

$$A_v = \frac{V_o}{V_i}$$

Convert:

$$V_i = 10\ \text{mV} = 0.01\ \text{V}$$

So,

$$A_v = \frac{1}{0.01} = 100$$

Answer: 100

9. If $V_i = 20\ \text{mV}$ and $V_o = 2\ \text{V}$, calculate A_v.

Convert:

$$V_i = 20\ \text{mV} = 0.02\ \text{V}$$

Then,

$$A_v = \frac{2}{0.02} = 100$$

Answer: 100

10. If $A_v = 50$, calculate gain in dB.

Formula:

$$\text{Gain(dB)} = 20\log_{10}(A_v)$$

$$\text{Gain(dB)} = 20\log_{10}(50)$$

$$\log_{10}(50) \approx 1.699$$

$$\text{Gain(dB)} = 20 \times 1.699 = 33.98 \text{ dB}$$

Answer: $\approx 34 \text{ dB}$

11. If $A_v = 100$, find gain in dB.

$$\text{Gain(dB)} = 20\log_{10}(100)$$

$$\log_{10}(100) = 2$$

$$\text{Gain(dB)} = 20 \times 2 = 40 \text{ dB}$$

Answer: 40 dB

12. If $V_i = 10 \text{ mV}$ and gain is 40, find V_o.

Formula:

$$A_v = \frac{V_o}{V_i} \Rightarrow V_o = A_v V_i$$

Convert:

$$V_i = 10 \text{ mV} = 0.01 \text{ V}$$

So,

$$V_o = 40 \times 0.01 = 0.4 \text{ V}$$

Answer: 0.4 V

13. If $V_o = 0.5 \text{ V}$ and $V_i = 10 \text{ mV}$, find gain in dB.

First find voltage gain:

$$A_v = \frac{V_o}{V_i} = \frac{0.5}{0.01} = 50$$

Now convert to dB:

$$\text{Gain(dB)} = 20\log_{10}(50)$$

$$= 20 \times 1.699 = 33.98 \text{ dB}$$

Answer: $\approx 34 \text{ dB}$

14. For $A_v = 10$, calculate gain in dB.

$$\text{Gain(dB)} = 20\log_{10}(10)$$
$$= 20 \times 1 = 20 \text{ dB}$$

Answer: 20 dB

15. For $A_v = 200$, calculate gain in dB.

$$\text{Gain(dB)} = 20\log_{10}(200)$$
$$\log_{10}(200) = 2.301$$
$$\text{Gain(dB)} = 20 \times 2.301 = 46.02 \text{ dB}$$

Answer: $\approx 46.02 \text{ dB}$

16. If input is 100 mV and output is inverted 3 V, find voltage gain with sign.

Formula:

$$A_v = \frac{V_o}{V_i}$$

Given:

$$V_i = 100 \text{ mV} = 0.1 \text{ V}$$
$$V_o = -3 \text{ V}$$

So,

$$A_v = \frac{-3}{0.1} = -30$$

Answer: -30

17. If $R_C = 10k\Omega$ and collector current changes by 0.2 mA, find change in collector voltage.

Formula:

$$\Delta V = \Delta I_C \times R_C$$

Convert:

$$\Delta I_C = 0.2 \text{ mA} = 0.0002 \text{ A}$$

Now,

$$\Delta V = 0.0002 \times 10000 = 2 \text{ V}$$

Answer: 2 V

18. If collector current increases from $1\ \mathrm{mA}$ to $1.5\ \mathrm{mA}$ with $R_C = 10k\Omega$, find change in voltage drop across R_C.

Change in current:

$$\Delta I_C = 1.5 - 1 = 0.5\ \mathrm{mA} = 0.0005\ \mathrm{A}$$

Now,

$$\Delta V_{RC} = \Delta I_C \times R_C$$

$$\Delta V_{RC} = 0.0005 \times 10000 = 5\ \mathrm{V}$$

Answer: $5\ \mathrm{V}$

19. If amplifier gain at midband is 40 dB, what is linear gain?

Formula:

$$\mathrm{Gain(dB)} = 20\log_{10}(A_v)$$

So,

$$40 = 20\log_{10}(A_v)$$

$$\log_{10}(A_v) = 2$$

$$A_v = 10^2 = 100$$

Answer: 100

20. If frequency increases and output drops from 2 V to 1 V for same input, what happens to gain in dB?

Since the **input remains the same**, voltage gain is proportional to output voltage.

Initial gain:

$$A_{v1} = \frac{2}{V_i}$$

Final gain:

$$A_{v2} = \frac{1}{V_i}$$

So the new gain is half of the old gain:

$$\frac{A_{v2}}{A_{v1}} = \frac{1}{2}$$

Change in gain in dB:

$$\Delta G = 20 \log_{10} \left(\frac{A_{v2}}{A_{v1}} \right)$$

$$\Delta G = 20 \log_{10} \left(\frac{1}{2} \right)$$

$$\Delta G = 20 \times (-0.3010) = -6.02 \text{ dB}$$

Answer:

The gain **decreases by approximately 6 dB**.

1. For a non-inverting amplifier, $R_f = 10k\Omega$, $R_i = 10k\Omega$. Find gain.

Formula:

$$A_v = 1 + \frac{R_f}{R_i}$$

Substitute:

$$A_v = 1 + \frac{10k}{10k} = 1 + 1 = 2$$

Answer: 2

2. If $V_{in} = 1\ \mathrm{V}$ and gain is 2, find V_{out}.

Formula:

$$V_{out} = A_v V_{in}$$

$$V_{out} = 2 \times 1 = 2\ \mathrm{V}$$

Answer: $2\ \mathrm{V}$

3. If $R_f = 20k\Omega$, $R_i = 10k\Omega$, find gain of non-inverting amplifier.

Formula:

$$A_v = 1 + \frac{R_f}{R_i}$$

$$A_v = 1 + \frac{20k}{10k} = 1 + 2 = 3$$

Answer: 3

4. For $V_{in} = 0.5\ \mathrm{V}$, $R_f = R_i$, find V_{out}.

If $R_f = R_i$, then for non-inverting amplifier:

$$A_v = 2$$

So,

$$V_{out} = A_v V_{in} = 2 \times 0.5 = 1\ \mathrm{V}$$

Answer: $1\ \mathrm{V}$

5. For an inverter, $R_f = 10k\Omega$, $R_i = 10k\Omega$. Find gain.

Formula for inverting amplifier:

$$A_v = -\frac{R_f}{R_i}$$

$$A_v = -\frac{10k}{10k} = -1$$

Answer: -1

6. If $V_{in} = 1\ \text{V}$ in an inverter, find V_{out}.

Formula:

$$V_{out} = A_v V_{in}$$

Since $A_v = -1$,

$$V_{out} = -1 \times 1 = -1\ \text{V}$$

Answer: $-1\ \text{V}$

7. For inverting amplifier, $R_i = 10k\Omega$, $R_f = 20k\Omega$. Find gain.

$$A_v = -\frac{R_f}{R_i}$$

$$A_v = -\frac{20k}{10k} = -2$$

Answer: -2

8. If $V_{in} = 1\ \text{V}$ and gain is -2, find V_{out}.

$$V_{out} = A_v V_{in}$$

$$V_{out} = (-2) \times 1 = -2\ \text{V}$$

Answer: $-2\ \text{V}$

9. If $V_{in} = 0.2\ \text{V}$, $R_f = 30k\Omega$, $R_i = 10k\Omega$, calculate V_{out}.

First find gain:

$$A_v = -\frac{R_f}{R_i} = -\frac{30k}{10k} = -3$$

Then:

$$V_{out} = A_v V_{in} = (-3) \times 0.2 = -0.6\ \text{V}$$

Answer: $-0.6\ \text{V}$

10. If $V_{out} = -4\ \text{V}$ and gain is -2, find V_{in}.

Formula:

$$V_{out} = A_v V_{in}$$

So,

$$V_{in} = \frac{V_{out}}{A_v}$$

$$V_{in} = \frac{-4}{-2} = 2\ \text{V}$$

Answer: $2\ \text{V}$

11. If LM741 is powered by $\pm 15\ \text{V}$, why can output clipping occur before $\pm 15\ \text{V}$?

The LM741 is not an ideal rail-to-rail op-amp. Its output cannot usually swing exactly up to the supply rails. Due to internal transistor voltage drops and circuit limitations, clipping often occurs a little before the supply limits.

Answer: Output clipping occurs before $\pm 15\ \text{V}$ because LM741 cannot swing exactly to the supply rails.

12. If output clips at $+13\ \text{V}$ and $-13\ \text{V}$, what is the maximum undistorted sine peak approximately?

The maximum undistorted sine amplitude must stay within the clipping levels.

So maximum peak value:

$$V_{peak(max)} \approx 13\ \text{V}$$

If needed, peak-to-peak value:

$$V_{pp(max)} \approx 26\ \text{V}$$

Answer: Maximum undistorted sine peak is approximately **13 V**.

13. In an inverting adder with equal resistors, $V_1 = 1\ \text{V}, V_2 = 0\ \text{V}$. Find V_{out}.

For equal resistors:

$$V_{out} = -(V_1 + V_2)$$

$$V_{out} = -(1 + 0) = -1\ \text{V}$$

Answer: $-1\ \text{V}$

14. If $V_1 = 1\ \text{V}, V_2 = +5\ \text{V}$, find V_{out}.

$$V_{out} = -(V_1 + V_2)$$

$$V_{out} = -(1 + 5) = -6\ \text{V}$$

Answer: $-6\ \text{V}$

15. If $V_1 = 1\text{ V}, V_2 = -5\text{ V}$, find V_{out}.

$$V_{out} = -(V_1 + V_2)$$

$$V_{out} = -(1 + (-5)) = -(-4) = 4\text{ V}$$

Answer: 4 V

16. If $V_1 = 2\sin\omega t$ and $V_2 = 3\text{ V}$, write the output expression for equal resistors.

For equal resistors in an inverting adder:

$$V_{out} = -(V_1 + V_2)$$

Substitute:

$$V_{out} = -(2\sin\omega t + 3)$$

So,

$$V_{out} = -2\sin\omega t - 3$$

Answer: $\boxed{V_{out} = -2\sin\omega t - 3}$

17. If $R_f = 10k\Omega, R_1 = 10k\Omega, R_2 = 20k\Omega, V_1 = 2\text{ V}, V_2 = 4\text{ V}$, find V_{out}.

Formula:

$$V_{out} = -R_f\left(\frac{V_1}{R_1} + \frac{V_2}{R_2}\right)$$

Substitute:

$$V_{out} = -10k\left(\frac{2}{10k} + \frac{4}{20k}\right)$$

$$V_{out} = -10k\left(\frac{2}{10k} + \frac{2}{10k}\right)$$

$$V_{out} = -10k\left(\frac{4}{10k}\right) = -4\text{ V}$$

Answer: -4 V

18. For an integrator, $R = 10k\Omega, C = 0.1\mu F$. Find RC.

Convert capacitance:

$$C = 0.1\mu F = 0.1 \times 10^{-6} F = 10^{-7} F$$

Now,

$$RC = 10k \times 0.1\mu F$$

$$RC = 10000 \times 10^{-7} = 10^{-3}\text{ s}$$

$$RC = 0.001\text{ s} = 1\text{ ms}$$

Answer: 1 ms

19. For the above integrator, find $\frac{1}{RC}$.

From previous answer:

$$RC = 0.001 \text{ s}$$

So,

$$\frac{1}{RC} = \frac{1}{0.001} = 1000 \text{ s}^{-1}$$

Answer: 1000 s^{-1}

20. Write the integrator equation:

$$V_{out} = -\frac{1}{RC} \int V_{in} \, dt$$

This is already the required equation.

Answer:

$$\boxed{V_{out} = -\frac{1}{RC} \int V_{in} \, dt}$$

21. If $V_{in} = 1 \text{ V (constant)}, R = 10k\Omega, C = 0.1\mu F$, find slope of output:

$$\frac{dV_{out}}{dt} = -\frac{V_{in}}{RC}$$

From above:

$$RC = 0.001 \text{ s}$$

Now:

$$\frac{dV_{out}}{dt} = -\frac{1}{0.001} = -1000 \text{ V/s}$$

Answer: -1000 V/s

22. For $R = 10k\Omega, C = 0.1\mu F$, and $V_{in} = 2 \text{ V}$, find output slope.

Formula:

$$\frac{dV_{out}}{dt} = -\frac{V_{in}}{RC}$$

$$\frac{dV_{out}}{dt} = -\frac{2}{0.001} = -2000 \text{ V/s}$$

Answer: -2000 V/s

23. If a square wave of ±1 V is applied to an integrator, what kind of output waveform is expected?

Integration of a square wave produces a triangular waveform.

Answer: A **triangular wave** is expected at the output.

24. What is the gain of a voltage follower?

A voltage follower has unity gain.

$$A_v = 1$$

Answer: 1

25. If $V_{in} = 2\sin\omega t$, write V_{out} for a buffer.

For a voltage follower:

$$V_{out} = V_{in}$$

So,

$$V_{out} = 2\sin\omega t$$

Answer: $\boxed{V_{out} = 2\sin\omega t}$

26. If input peak is 2 V, what is output peak in a voltage follower?

Since gain is 1:

$$V_{out(peak)} = V_{in(peak)} = 2\ \mathrm{V}$$

Answer: $2\ \mathrm{V}$

27. If $V_{in} = 1\ \mathrm{V}_{p-p}$, what is V_{out} for an ideal voltage follower?

For an ideal voltage follower:

$$V_{out} = V_{in}$$

So output remains:

$$1\ \mathrm{V}_{p-p}$$

Answer: $1\ \mathrm{V}_{p-p}$